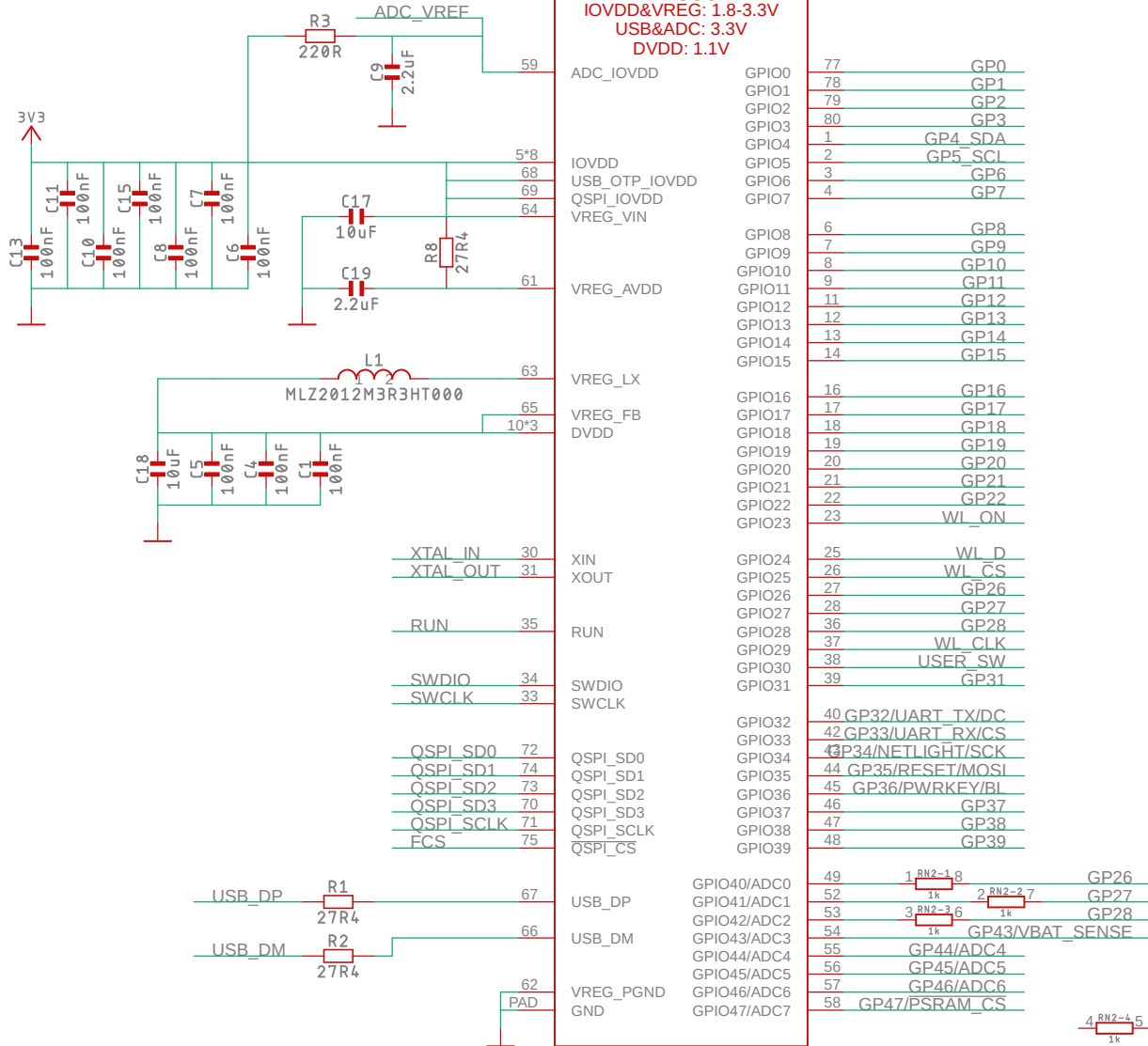


Pimoroni Pico LiPo 2 XL W (PIM776)

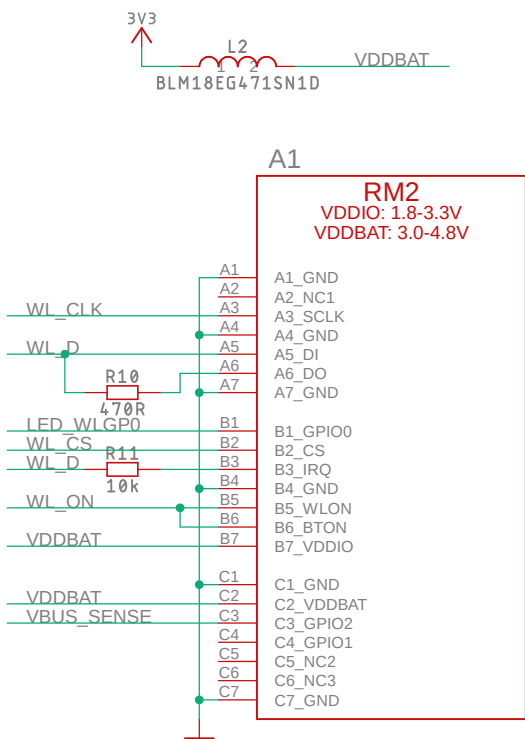
RP2350B MCU

MCU2

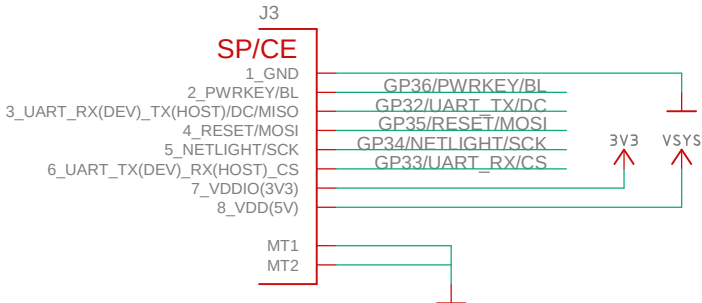
RP2350B
IOVDD&VREG: 1.8-3.3V
USB&ADC: 3.3V
DVDD: 1.1V



RM2 module



SP/CE Connector



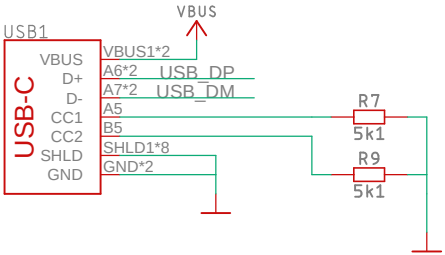
PIMORONI

DATE GENERATED SHEET

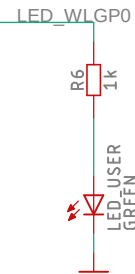
11/07/2025 17:20 1/4

Pimoroni Pico LiPo 2 XL W (PIM776)

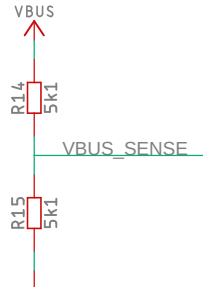
USB-C connector



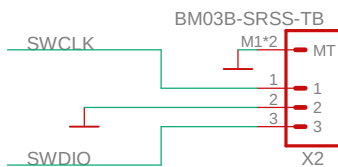
User LED



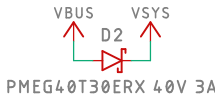
Vbus sense voltage divisor



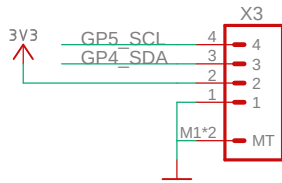
Debug connector



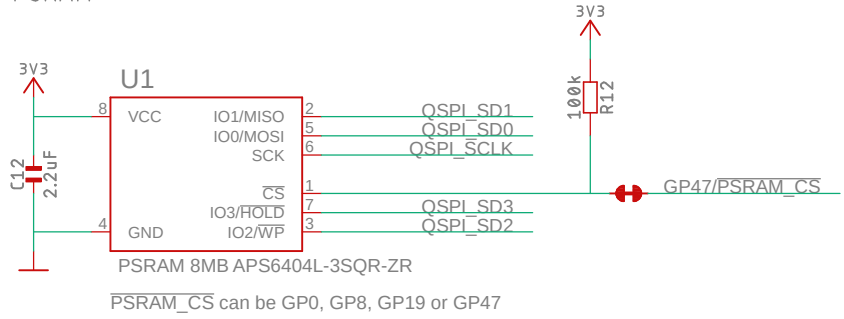
Vbus to Vsys diode connection



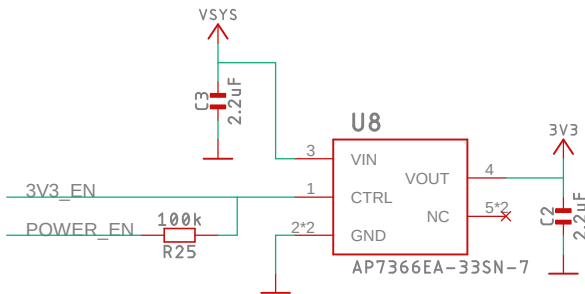
QW/ST connector



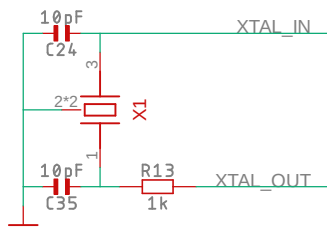
PSRAM



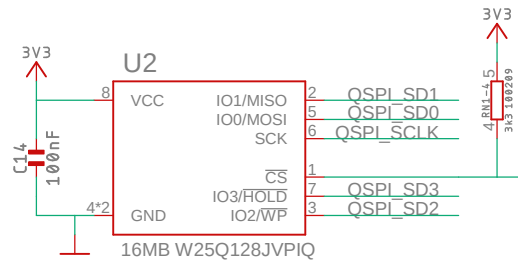
LDO for 3v3 output, low Iq



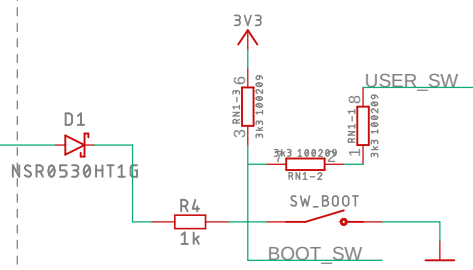
Crystal oscillator



Flash memory

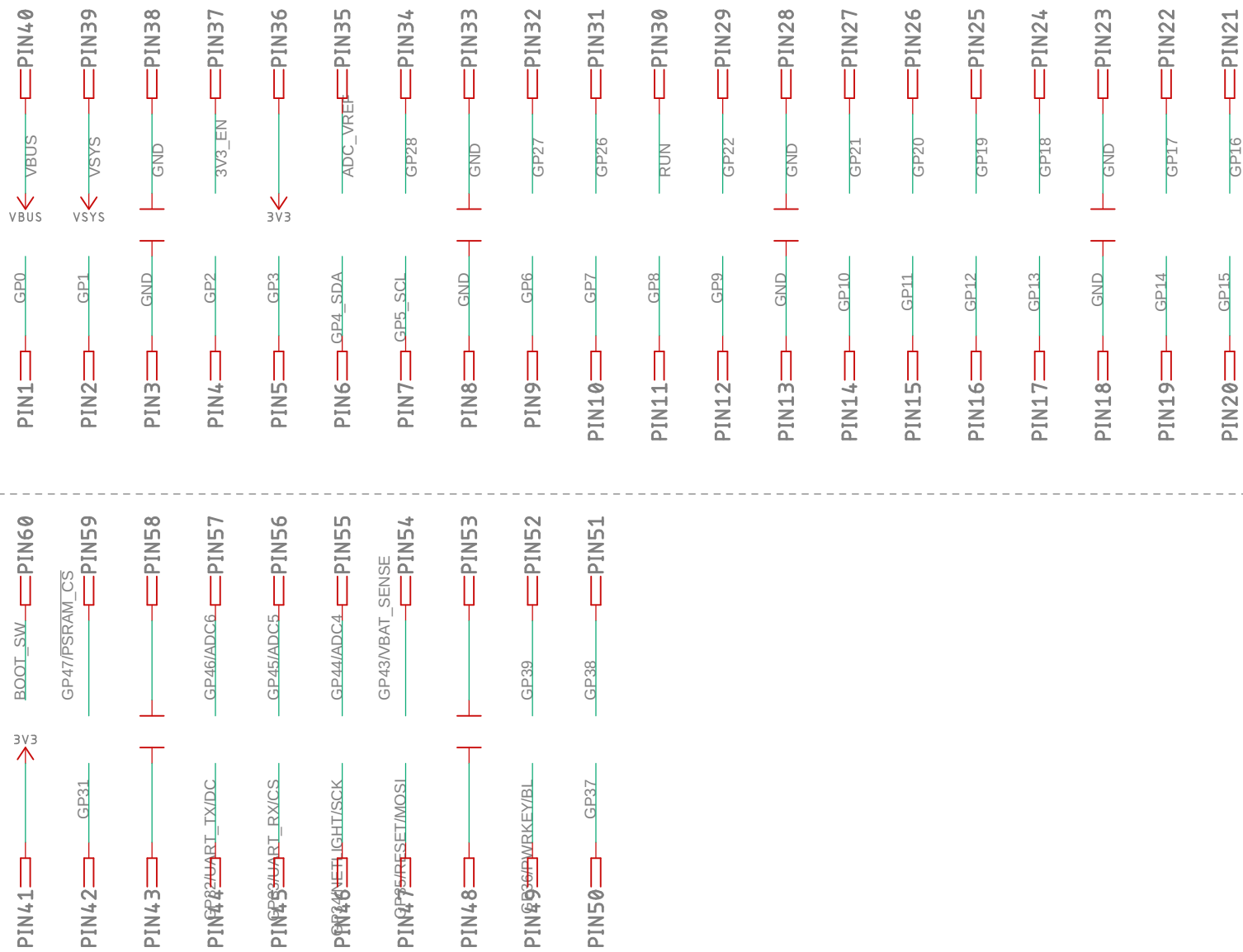


Combined USB boot & user switch



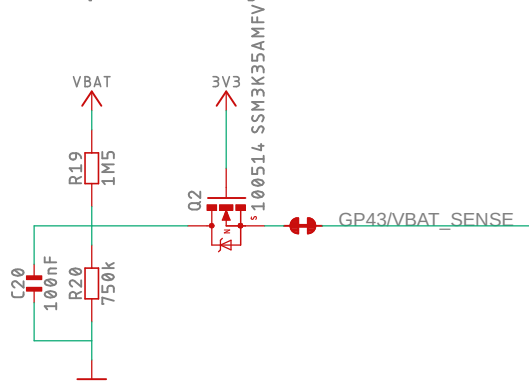
Pimoroni Pico LiPo 2 XL W (PIM776)

Output headers



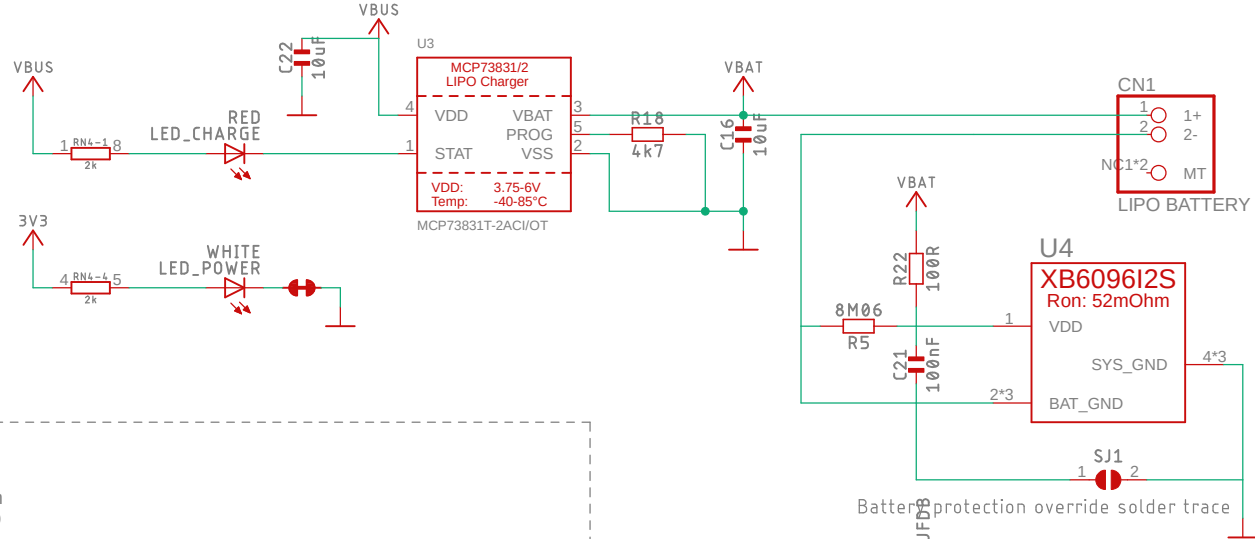
Pimoroni Pico LiPo 2 XL W (PIM776)

Battery sense voltage divisor



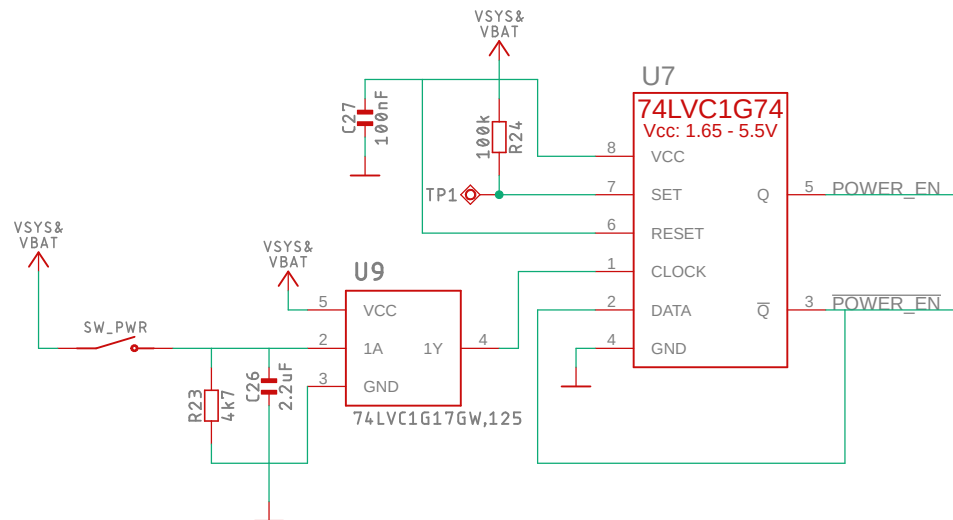
Lipo charger, protection and connector

4.7k on PROG sets charge current limit to ~220mA.



Power switch toggle circuit

R24 and the PCB capacitor slow down the SET rising slope enough compared to RESET that the chip reliably defaults to on at powerup



Vbat charge protection and power switch FET switches

